

Security Control Verification Matrix

The following matrix summarises the security and integration controls verified for the TEAVS–ADCRS core integration endpoint:

POST /api/ingestion/core

Control Area	Test Performed	Expected Result	Actual Result	Status	Evidence
API Availability	API Gateway health check	HTTP 200	HTTP 200	PASS	EVID-AUTO-01
Authentication	Request without JWT	HTTP 401	HTTP 401 – No token provided	PASS	EVID-04
Authentication	Request with invalid JWT	HTTP 401	HTTP 401 – Invalid token	PASS	EVID-05
RBAC	Valid JWT with analyst role	HTTP 403	HTTP 403 – Access denied	PASS	EVID-07
RBAC	Valid JWT with ingestion_service role	Request permitted	HTTP 202 Accepted	PASS	EVID-09
Input Validation	Empty/incomplete payload	HTTP 400	HTTP 400	PASS	EVID-AUTO-01
Input Validation	Incorrect field types	HTTP 400	HTTP 400	PASS	EVID-AUTO-01
JSON Parsing	Malformed JSON	HTTP 400	HTTP 400	PASS	EVID-AUTO-01
Automated Regression	Automated security test suite	All defined tests pass	8/8 tests passed	PASS	EVID-AUTO-01
Rate Limiting	Burst of repeated	Requests above threshold rejected	HTTP 429 returned after threshold	PASS	EVID-RATE-01

	valid requests				
Duplicate Handling	Same valid payload submitted twice	Duplicate should ideally be identified	Both requests accepted with HTTP 202	FINDING	EVID-DUP-01
Dependency Failure	RabbitMQ deliberately stopped	Request should not be reported as accepted	HTTP 500 returned	PASS	EVID-FAIL-01
RabbitMQ Delivery	Valid TEAVS E2E payload submitted	Message reaches integration queue	Message received from core_model_integration_queue	PASS	EVID-E2E-02
Downstream Processing	RabbitMQ message consumed	Payload reaches coreModelIntegration()	Processing function received payload	PASS	EVID-E2E-03
Model Dependency	Core processing attempts model load	Trained model available	Training model not found	BLOCKED	EVID-E2E-04
Model Asset Availability	Repository checked for configured model file	Model artifact present	No .joblib, .pkl, or .pickle model found	BLOCKED	EVID-E2E-05

Overall Control Assessment

The testing confirmed that the main API security controls around the TEAVS-ADCRS integration are working as expected.

JWT authentication successfully blocks unauthenticated and invalid-token requests. RBAC now restricts the core ingestion endpoint to approved roles, and input validation prevents incomplete or structurally invalid payloads from being accepted.

Rate limiting was also added and verified to prevent excessive repeated requests from a single client.

The automated tester provides repeatable verification for the main authentication, authorisation and input-validation controls, with all eight implemented checks passing after remediation.

Two areas remain outside the fully verified state.

The first is duplicate handling. Identical TEAVS payloads can currently be accepted more than once, and no idempotency control was identified in the inspected gateway, RabbitMQ consumer or core-model processing path.

The second is final model inference. End-to-end testing confirmed successful transport of a TEAVS payload through the API Gateway, RabbitMQ and data-ingestion-service into `coreModelIntegration()`. Full inference could not be completed because the configured trained model artifact was not available in the local project environment.

These remaining items are documented as integration dependencies and follow-up areas rather than failures of the implemented authentication, RBAC, validation or messaging controls.